

Sequential Algorithms

Learning Objectives

- Identify and construct sequential algorithms in real contexts
- Understand how step order affects algorithm outcomes
- Design sequential solutions for multi-step computational problems

Engage (5-7 minutes)

Present two recipes for the same dish but with steps in different orders. Ask students: does order matter? Then show a sorting algorithm where swapping two steps produces incorrect results. Establish that sequential algorithms depend critically on step ordering.

Explore (10-12 minutes)

Students design a sequential algorithm for a task with multiple dependencies: planning a school event (book venue → send invitations → order catering → set up decorations → welcome guests). They must identify which steps can happen in parallel and which must be sequential, then write the complete ordered procedure.

Explain (10-12 minutes)

A sequential algorithm executes instructions one after another in a fixed order, with each step depending on the completion of the previous one. This is the simplest control flow structure. The key principle is that changing the order of steps changes the result—operations must respect data dependencies. Sequential algorithms form the backbone of all programs; even complex algorithms decompose into sequential blocks at the lowest level.

Elaborate (8-10 minutes)

Analyze sequential algorithms in computation: the fetch-decode-execute cycle of a CPU, the steps in network packet transmission, or the sequence of operations in database queries. Discuss how modern processors use pipelining to overlap sequential steps for efficiency. Examine a real algorithm (e.g., Euclidean algorithm for GCD) and trace each sequential step with concrete values.

Evaluate (5-8 minutes)

Students receive a shuffled list of steps for an algorithm (e.g., binary conversion) and must order them correctly. Then they write their own sequential algorithm for a novel problem (calculating tips

and splitting a bill among friends). Peer evaluation focuses on step completeness and correct ordering.